Experiment -10

Aim:

To simulate routing information Protocol (RIP) using Cisco Packet tracer and observe dynamic routing between different networks

IP Configuration:

Device	Interface	IP address
PC0	Fast Ethernet	
Router 0	Fa 0/1	10.0.0.218
Router 0	S0/0/1	10.0.0.1/8
Router 1	S0/0/0	192.168.1.254/30
Router 1	S0/0/1	192.168.1.250/30
Router 2	S0/0/0	192.168.1.249/
Router 2	S0/0/1	192.668.1.253/
PC1	Fast Ethernet	20.0.0.218

Procedure

1) Create the topology with 3 routers and 2PCs
2) Connect the routers using serial cables and connect PC's to the respective routers.
3) Assign IP address to all PCs and router interface
4) Enable the router interface.

5) Configure the remaining interface according to the given IP Table

6) Enable RIP on router 0 and configure RIP on routers 1 and 2.

7) From PC0, use Ping 20.0.0.2 to verify connectivity

8) Use traceroute to observe the route taken by packets

9) Disconnect the direct Router 0 - Router 2 link and observe that RIP routes the traffic through Router 1

Router1

```
Router>enable
Router#configure
terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/1
Router(config-if)#ip address 192.168.1.246 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth
64          Router(config-if)#no
shutdown        Router(config-
```

Router2

```
Router>enable
Router#configure
terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip address 20.0.0.1
255.0.0.0 Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 192.168.1.245
255.255.255.252 Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/1
Router(config-if)#ip address 192.168.1.253
255.255.255.252 Router(config-if)#no shutdown
```

```
Router>enable
Router#configure
terminal
Enter configuration commands, one per line. End with CNTL/Z
Router(config)#interface fastEthernet 0/0
Router(config-if)#ip address 10.0.0.1
255.0.0.0 Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 2000000
[Output omitted]
```

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 192.168.1.249 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth 64
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface serial 0/0/1
Router(config-if)#ip address 192.168.1.254 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth
64 Router(config-if)#no
shutdown Router(config-
```

```
Router0(config)#router rip
Router0(config-router)# network
10.0.0.0
Router0(config-router)# network 192.168.1.252
```

Router0

Router1

```
Router1(config)#router rip
Router1(config-router)# network 192.168.1.244
Router1(config-router)# network 192.168.1.248
```

Router2

```
Router2(config)#router rip
```

Result :

Thus RIP, was successfully simulated in cisco Packet Tracer and dynamic route selection and routing were observed.